Radar and Defense System

Components

- Servo Motor x2
- Ultrasonic sensor
- Arduino Uno/esp32
- Half sized BreadBoard
- LCD display
- Buzzer
- Some jumper wires

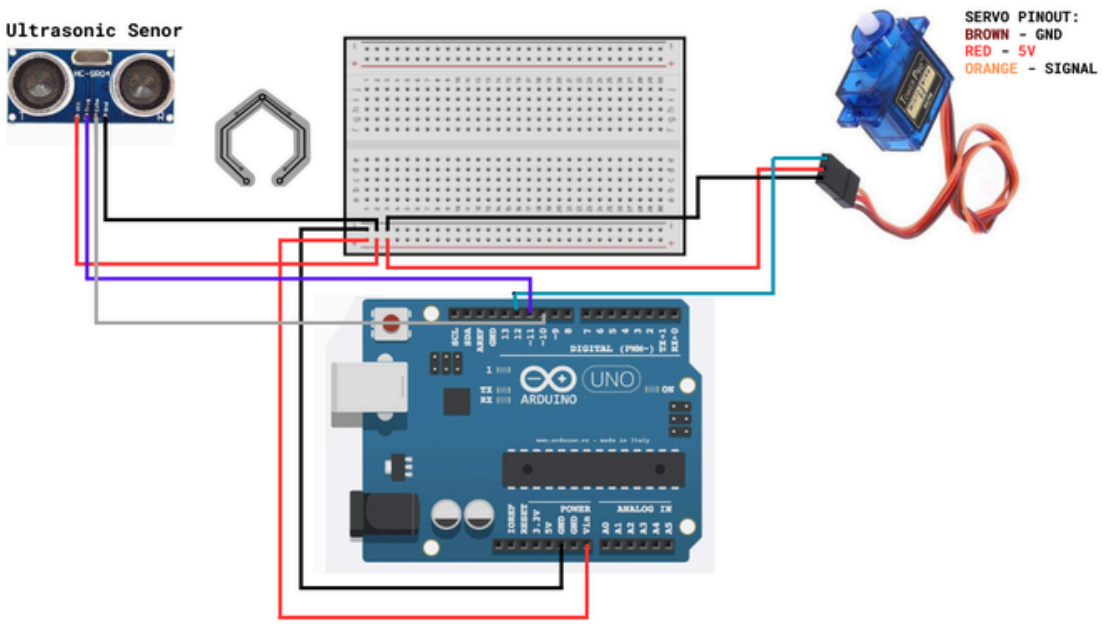
Theory

The proposed system is an **ESP32-based Radar and Defence Prototype** designed to simulate how surveillance and automated defence mechanisms work. The system uses an **ultrasonic (US) sensor** to detect objects (representing enemy targets) and activates a **defence response** when a target is identified within a defined range.

Project Overview

A **smart water dispenser** that:

1. The **ultrasonic sensor** emits high-frequency sound waves.
2. Echoes reflected from objects are received back.
3. The **ESP32 calculates distance** using time-of-flight.
4. A **servo motor rotates the sensor**, scanning a defined angular range (e.g., 0°–180°).
5. Each scan cycle maps **angle + distance** data.
6. Object presence is interpreted as a **potential target (enemy)**.



Code (just radar) (IDE)

```
#include <Servo.h>

// Ultrasonic Sensor Pins
const int trigPin = 10;
const int echoPin = 11;

// Buzzer Pin
const int buzzerPin = 3;

// Variables
long duration;
int distance;

// Detection threshold (cm)
const int threshold = 50;

Servo myServo;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, OUTPUT);

  Serial.begin(9600);
  myServo.attach(12);
}

void loop() {

  // Scan from 15° → 165°
  for(int i = 15; i <= 165; i++) {
    myServo.write(i);
    delay(30);

    distance = calculateDistance();

    controlBuzzer(distance);

    Serial.print(i);
    Serial.print(",");
    Serial.print(distance);
    Serial.print(".");
  }

  // Scan from 165° → 15°
```

```
  for(int i = 165; i > 15; i--) {
    myServo.write(i);
    delay(30);

    distance = calculateDistance();

    controlBuzzer(distance);

    Serial.print(i);
    Serial.print(",");
    Serial.print(distance);
    Serial.print(".");
  }
}

// Distance calculation
int calculateDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

  distance = duration * 0.034 / 2;
  return distance;
}

// Buzzer control logic
void controlBuzzer(int dist) {
  if(dist > 0 && dist < threshold) {
    digitalWrite(buzzerPin, HIGH);  // Buzzer ON
  }
  else {
    digitalWrite(buzzerPin, LOW);   // Buzzer OFF
  }
}
```

Code (Processing)

```
import processing.serial.*; // imports library for serial communication
import java.awt.event.KeyEvent; // imports library for reading the data
from the serial port
import java.io.IOException;
Serial myPort; // defines Object Serial
```

```
// defubes variables
String angle="";
String distance="";
String data="";
String noObject;
float pixsDistance;
int iAngle, iDistance;
int index1=0;
int index2=0;
PFont orcFont;
void setup() {

 size (1200, 700); // ***CHANGE THIS TO YOUR SCREEN RESOLUTION***
 smooth();
 myPort = new Serial(this,"COM5", 9600); // starts the serial
communication
 myPort.bufferUntil('.'); // reads the data from the serial port up to
the character '.'. So actually it reads this: angle,distance.
}
void draw() {

  fill(98,245,31);
  // simulating motion blur and slow fade of the moving line
  noStroke();
  fill(0,4);
  rect(0, 0, width, height-height*0.065);

  fill(98,245,31); // green color
  // calls the functions for drawing the radar
  drawRadar();
  drawLine();
  drawObject();
  drawText();
}
void serialEvent (Serial myPort) { // starts reading data from the
Serial Port
  // reads the data from the Serial Port up to the character '.' and
puts it into the String variable "data".
  data = myPort.readStringUntil('.');
  data = data.substring(0,data.length()-1);

  index1 = data.indexOf(","); // find the character ',' and puts it
into the variable "index1"
  angle= data.substring(0, index1); // read the data from position "0"
to position of the variable index1 or thats the value of the angle the
Arduino Board sent into the Serial Port
  distance= data.substring(index1+1, data.length()); // read the data
from position "index1" to the end of the data pr thats the value of the
distance

  // converts the String variables into Integer
```

```
  iAngle = int(angle);
  iDistance = int(distance);
}
void drawRadar() {
  pushMatrix();
  translate(width/2,height-height*0.074); // moves the starting
coordinats to new location
  noFill();
  strokeWeight(2);
  stroke(98,245,31);
  // draws the arc lines
  arc(0,0,(width-width*0.0625),(width-width*0.0625),PI,TWO_PI);
  arc(0,0,(width-width*0.27),(width-width*0.27),PI,TWO_PI);
  arc(0,0,(width-width*0.479),(width-width*0.479),PI,TWO_PI);
  arc(0,0,(width-width*0.687),(width-width*0.687),PI,TWO_PI);
  // draws the angle lines
  line(-width/2,0,width/2,0);
  line(0,0,(-width/2)*cos(radians(30)),(-width/2)*sin(radians(30)));
  line(0,0,(-width/2)*cos(radians(60)),(-width/2)*sin(radians(60)));
  line(0,0,(-width/2)*cos(radians(90)),(-width/2)*sin(radians(90)));
  line(0,0,(-width/2)*cos(radians(120)),(-width/2)*sin(radians(120)));
  line(0,0,(-width/2)*cos(radians(150)),(-width/2)*sin(radians(150)));
  line((-width/2)*cos(radians(30)),0,width/2,0);
  popMatrix();
}
void drawObject() {
  pushMatrix();
  translate(width/2,height-height*0.074); // moves the starting
coordinats to new location
  strokeWeight(9);
  stroke(255,10,10); // red color
  pixsDistance = iDistance*((height-height*0.1666)*0.025); // covers
the distance from the sensor from cm to pixels
  // limiting the range to 40 cms
  if(iDistance<40){
    // draws the object according to the angle and the distance

line(pixsDistance*cos(radians(iAngle)),-pixsDistance*sin(radians(iAngle
)),(width-width*0.505)*cos(radians(iAngle)),-(width-width*0.505)*sin(ra
dians(iAngle)));
  }
  popMatrix();
}
void drawLine() {
  pushMatrix();
  strokeWeight(9);
  stroke(30,250,60);
  translate(width/2,height-height*0.074); // moves the starting
coordinats to new location
```

```
line(0,0,(height-height*0.12)*cos(radians(iAngle)),-(height-height*0.12
)*sin(radians(iAngle))); // draws the line according to the angle
  popMatrix();
}
void drawText() { // draws the texts on the screen

  pushMatrix();
  if(iDistance>40) {
  noObject = "Out of Range";
  }
  else {
  noObject = "In Range";
  }
  fill(0,0,0);
  noStroke();
  rect(0, height-height*0.0648, width, height);
  fill(98,245,31);
  textSize(25);

  text("10cm",width-width*0.3854,height-height*0.0833);
  text("20cm",width-width*0.281,height-height*0.0833);
  text("30cm",width-width*0.177,height-height*0.0833);
  text("40cm",width-width*0.0729,height-height*0.0833);
  textSize(40);
  text("SciCraft ", width-width*0.875, height-height*0.0277);
  text("Angle: " + iAngle +" °", width-width*0.48,
height-height*0.0277);
  text("Distance: ", width-width*0.26, height-height*0.0277);
  if(iDistance<40) {
  text("        " + iDistance +" cm", width-width*0.225,
height-height*0.0277);
  }
  textSize(25);
  fill(98,245,60);

translate((width-width*0.4994)+width/2*cos(radians(30)),(height-height*
0.0907)-width/2*sin(radians(30)));
  rotate(-radians(-60));
  text("30°",0,0);
  resetMatrix();

translate((width-width*0.503)+width/2*cos(radians(60)),(height-height*0
.0888)-width/2*sin(radians(60)));
  rotate(-radians(-30));
  text("60°",0,0);
  resetMatrix();

translate((width-width*0.507)+width/2*cos(radians(90)),(height-height*0
.0833)-width/2*sin(radians(90)));
  rotate(radians(0));
```

```
  text("90°",0,0);
  resetMatrix();

translate(width-width*0.513+width/2*cos(radians(120)),(height-height*0.
07129)-width/2*sin(radians(120)));
  rotate(radians(-30));
  text("120°",0,0);
  resetMatrix();

translate((width-width*0.5104)+width/2*cos(radians(150)),(height-height
*0.0574)-width/2*sin(radians(150)));
  rotate(radians(-60));
  text("150°",0,0);
  popMatrix();
}
```

Connections

📡 Ultrasonic Sensor (HC-SR04)

HC-SR04 Pin	Connect To Arduino Nano
VCC	5V
GND	GND
TRIG	D10
ECHO	D11

Buzzer

Buzzer Pin	Connect To Arduino Nano
+ (Positive)	D3
– (Negative)	GND

Servo Motor

Servo Wire	Connect To Arduino Nano
Signal (Orange/Yellow)	D12
VCC (Red)	5V *(or external 5V recommended)*

GND (Brown/Black) GND